23.4.2020

SESSION – XII (B) p5–11

(Trivedi) HYPOTHESIS TESTING

Statistical Inference
Main Problems

Estimation of Parameters | Tests of statistical hypothesis

Statistical Hypothesis – A statement or assertion about one or more probability distribution associated with population

Statistical Tests Procedures to decide whether to reject or accept hypothesis based on information contained in a sample. $x_1, x_2, \ldots, x_n$ from a population distribution F_θ.

↓

Functional form is known except parameter θ.

Examples

1. Coin Tossing – whether prob. p of head equals p_0.
2. System Administrator – to decide to upgrade the capacity of the installation.
3. Battery life – T hours (lasts)

We need to take a random sample to test the hypothesis

We form <u>Null hypothesis</u> (claim about prob. distn)

<u>Alternative hypothesis</u> (Contradicting hypothesis)

H_0 (Null hypothesis) vs H_1 (Alternative hypothesis)

<u>Example 1</u>: MTTF (mean time to failure)

H_0: MTTF exceeds say θ_0 hours

H_1: MTTF $< \theta_0$

<u>Example 2</u> Manufacturing process of certain electronic components. have average life μ hrs

-> It is proposed to test a new procedure for manufacturing of electronic component!

H_0: $\mu = \mu_0$

H_1: $\mu > \mu_0$ or $\mu < \mu_0$ or $\mu \neq \mu_0$

From Rohatgi & Saleh) <u>Defn</u> <u>Parametric Hypothesis</u>

— is an assertion about the unknown parameter θ. It is referred to as <u>Null hypothesis</u> $H_0: \theta \in \Theta_0 \subset \Theta$,

— the statement $H_1: \theta \in \Theta_1 = \Theta - \Theta_0$ is usually referred to as alternative hypothesis H_1

Random sample $X_1, X_2, \ldots X_n$.

X_i is a rv for each i

R^n (: the totality of all n-tuples will span Euclidean n-space R^n).

Divide n-space R^n into two regions

(Acceptance Region) $R(H_0)$ $\qquad$ $R(H_1)$ (Critical or Rejection region)

<u>Decision Rule</u> : If sample point $\underset{\sim}{x} = (x_1, x_2, \ldots, x_n)$ lies in $R(H_1)$, we reject H_0

<u>On the other hand</u> If the observed n-tuple lies in $R(H_0)$, we fail* to reject H_0

* <u>Note</u>: The sample does not have enough evidence against H_0

<u>Two Types of Errors</u>

Type I error – can be made – may reject H_0 when true

Type II error – Accept H_0 when false

	True: H_0	True: H_1
H_0	Correct	Type II Error
Accept H_1	Type I error	Correct

$P(\text{Type I error}) = \alpha$

$\downarrow$

Ho True but sample lies in rejection region (leading us to reject Ho)

α is known as Level of significance of the test

$P(\text{Type II error}) = \beta$

Ho false but sample lies in Acceptance Region

$(1-\beta)$ is known as power of the test.

<u>Summary</u>

$$\alpha = P(\text{Type I error})$$
$$= P(\text{Rejecting Ho when it is true})$$

$$\beta = P(\text{Type II error})$$
$$= P(\text{Accepting Ho when it is False})$$

<u>Wrong Decision</u> : Type I or Type II errors lead to Wrong Decision

— Attempt to minimize these errors

<u>Fix sample size</u> : a decrease in one type error leads to an increase in other type of error.

1. If C is the critical region, then $P_\theta C$ $\theta \in \Theta_0$, (Prob of Type I error)

where $C = \{ \underset{\sim}{x} \in \mathbb{R}^n : H_0 \text{ rejected if } \underset{\sim}{x} \in C \}$

2. $P_\theta C^c$, $\theta \in \Theta_1$ is (Prob: Type II error)

Partition the sample space into two disjoint sets C and C^c

<u>Ideal Situation</u>: Find critical region for which these probabilities are zero.

— Such situations do not arise in practice

<u>Procedure in Practice</u>:

Rohatgi & Saleh

Limit Type I error to a prearssigned level α say (0.01 or 0.05)

Minimize prob. of Type II error.

Tests on the Population Mean

Simple Hypothesis $H_0 : \theta = \theta_0$ (parameter θ of the population)

<u>Steps</u>:

1. Find statistic $W = W(X_1, X_2, \cdots X_n)$ such that distribution of W is known.
2. Choose an interval (a, b) such that $P(W \notin (a,b) \mid H_0 \text{ is true}) = \alpha$

We can write

$$P(a < W < b \mid H_0 \text{ true}) = 1-\alpha$$

i.e.

$$P(a < W(x_1, \ldots x_n) < b) = 1-\alpha$$

3. Take a sample $x_1, \ldots x_n$ and compute $w = W(x_1, \ldots x_n)$
if $w \notin (a,b)$ reject H_0 in favour of H_1;
otherwise fail to reject H_0

<u>Problem</u> (Trivial) Test a hypothesis regarding population mean μ based on a random sample of size n from a normal population with known variance.

$$H_0 : \mu = \mu_0$$
$$H_1 : \mu \neq \mu_0$$

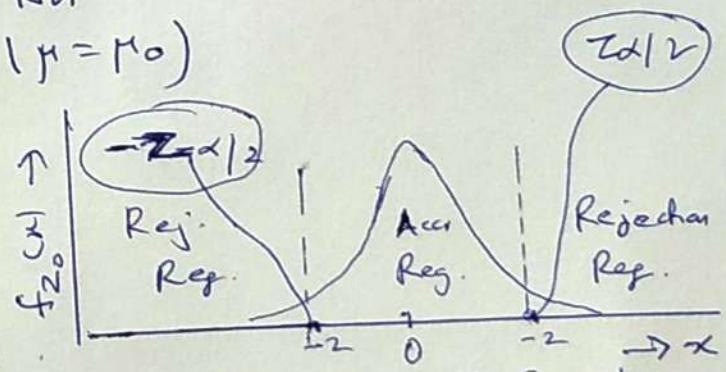
Test Statistic $\quad Z = \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}} \qquad \underline{(1)}$

Note $Z \sim N(0,1)$.

Let $\quad Z_0 = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \qquad \underline{(2)}$

$$\alpha = 1 - P(-b < z < b \mid \mu = \mu_0)$$

$$b = z_{\alpha/2} \qquad \underline{(3)}$$



(Acceptance & Rejection Regions)

<u>Acceptance Region</u>

$$-z_{\alpha/2} < Z_0 < z_{\alpha/2} \qquad \underline{(4)}$$

or $\quad \mu_0 - \dfrac{z_{\alpha/2}\sigma}{\sqrt{n}} < \bar{X} < \mu_0 + \dfrac{z_{\alpha/2}\sigma}{\sqrt{n}}$.

Rejection region (5)

$$|\bar{x} - \mu_0| > \frac{z_{\alpha/2}\,\sigma}{\sqrt{n}}$$

Ex. (Trivedi)

Program's average working set size

$\mu_0 = 50$ pages $\sigma^2 = 900$ pages^2

Reorganization decreased working set size

Test the hypothesis

$H_0: \mu = \mu_0$ versus $H_1: \mu < \mu_0$

$Z_\alpha = Z_{0.05} = 1.645$ at 5% level of significance

The critical region

$$\bar{x} < \mu_0 - \frac{\sigma}{\sqrt{n}} Z_\alpha = 50 - \frac{30}{\sqrt{n}} \times 1.645$$

or $\bar{x} < 50 - \frac{49.35}{\sqrt{n}}$

Take samples $n = 100$

If $\bar{x} < 45$, we have reason to believe reorganization improved program locality

SESSION XIII

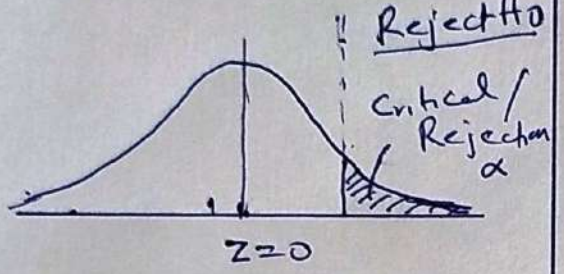
One tailed and Two tailed Tests

$H_0 : \mu = \mu_0$

$H_1 : \mu > \mu_0$

(Right Tailed)

The critical region on the right



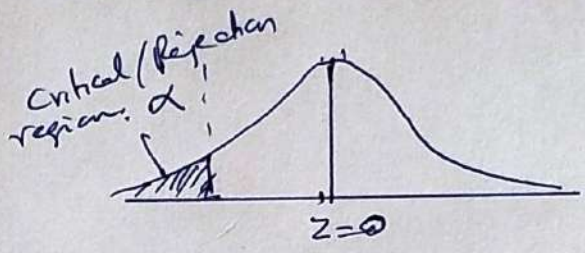
$P(Z > z_\alpha) = \alpha$

$H_0 : \mu = \mu_0$

$H_1 : \mu < \mu_0$

(Left Tailed)

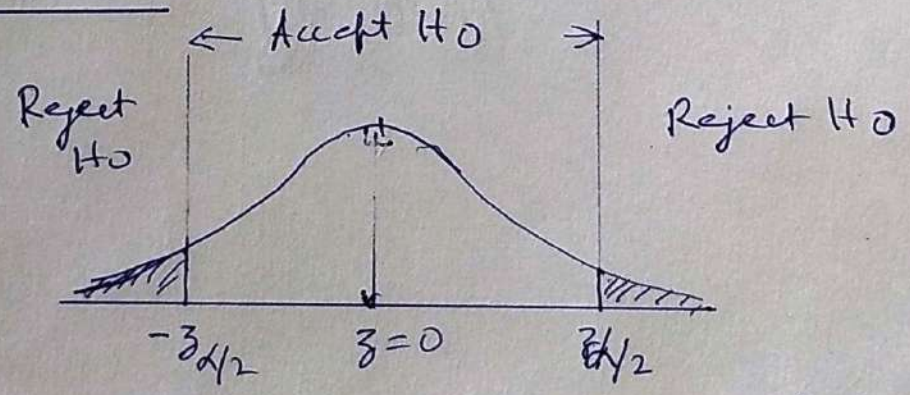
The critical region on the left



$P(Z < -z_\alpha) = \alpha$

Prob. α that a random value of the test statistic lies in the critical region represents level of significance.

Two tailed Test



Null hyp. $H_0 : \mu = \mu_0$; Alternative hypothesis $H_1 : \mu \neq \mu_0$

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Sample Sizes Needed

Importance of sample size

Both Type I error (α) and Type II error (β) Decrease as n increases.

Allows us to give desired power in hypothesis test

(From Arnold A Allen)

Two - Sample Tests of Means

Comparing the performance of two entities

e.g. - productivity of two groups.

- difference in effectiveness of two
- software systems or programming languages
- difference in down times of two systems.

We are requd to take samples from two populations.

NOTE: Suppose two independent rvs have means μ_X and μ_Y and variances σ_X^2 and σ_Y^2. Then the distribution of $X-Y$ has the mean $\mu_X - \mu_Y$ and variance $\sigma_X^2 + \sigma_Y^2$.

A random sample of size n_1 is taken from the population determined by X and a random sample of size n_2 is taken from the population determined by Y, then it can be shown

$$E[\bar{X} - \bar{Y}] = \mu_X - \mu_Y$$

$$var[\bar{X} - \bar{Y}] = \frac{\sigma_X^2}{n_1} + \frac{\sigma_Y^2}{n_2}$$

The statistic

$$Z = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{[Var(\bar{X} - \bar{Y})]^{1/2}}$$

has standard normal distribution.

<u>Critical Regions for Test statistic Z</u>

$Z < -z_\alpha$ if $H_1: \mu_X - \mu_Y < d_0$

$Z > +z_\alpha$ if $H_1: \mu_X - \mu_Y > d_0$

$Z < -z_{\alpha/2}$ or $Z > z_{\alpha/2}$ if $H_1: \mu_X - \mu_Y \neq d_0$

The null hypothesis

$H_0: \mu_X - \mu_Y = d_0$

(Difference between the two population means equals some given value).

Ex.

Two file servers are compared according to their response time for retrieving a small file. The mean response time of 50 such requests submitted to

Server 1	Server 2
Mean 682 ms	Mean: 675 ms
s.d. 25 ms	s.d 28 ms.

To test the hypothesis that server 2 has better response than server 1, we form

$H_0: \mu_X = \mu_Y$ (no difference in response time)

$H_1: \mu_X > \mu_Y$ (server 2 is better than server 1)

Null hypothesis

$\mu_X - \mu_Y = 0$

$$\sigma_{\bar{X}-\bar{Y}} = \sqrt{\frac{\sigma_X^2}{n_1} + \frac{\sigma_Y^2}{n_2}} = \sqrt{\frac{(25)^2}{50} + \frac{(28)^2}{50}}$$

$$= 5.3$$

Test statistic

$$Z = \frac{\bar{X} - \bar{Y}}{\sigma_{\bar{X}-\bar{Y}}} = \frac{\bar{X} - \bar{Y}}{5.3}$$

Value of statistic

$$z = \frac{682 - 675}{5.3} = 1.32$$

Use one tailed test at 5% level of significance

$$Z_\alpha = Z_{0.05} = 1.645$$

Since calculated value 1.32 is less than critical value 1.645; we fail to reject H_0

We cannot support the claim server 2 is more responsive than server 1.

Large sample theory :- application of normal distribution (central limit theorem)

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

It may be noted that for small sample size, the test based on normality cannot be applied. Test based on students' t-distribution.

<u>Students t-distribution</u> Given a random sample $x_1, x_2, \ldots x_n$ from normal population with mean μ and σ^2, the test statistic is

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}, \quad \bar{x} = \frac{\sum x_i}{n}, \quad s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

This follows Students t-distribution with $\nu = (n-1)$ df. The pdf is

$$f(t) = \frac{1}{\sqrt{\nu}\, B\left(\frac{1}{2}, \frac{\nu}{2}\right)} \frac{1}{\left(1 + \frac{t^2}{\nu}\right)^{\frac{\nu+1}{2}}}$$

<u>Note</u> :- Both the statistic and the sampling distribution are independent of σ^2.

For large $\nu = (n-1)$, the distribution of t is asymptotically normal.

Small (exact) <u>sample theory</u> :

From Allen

Recall Procedure for testing a hypothesis

Step 1 Formulate H_0 and H_1.

Usually H_1 is the hypothesis we want to test.

Step 2 select appropriate statistic say

$(X_1, X_2, \cdots X_n)$

e.g sample mean $\bar{X}$

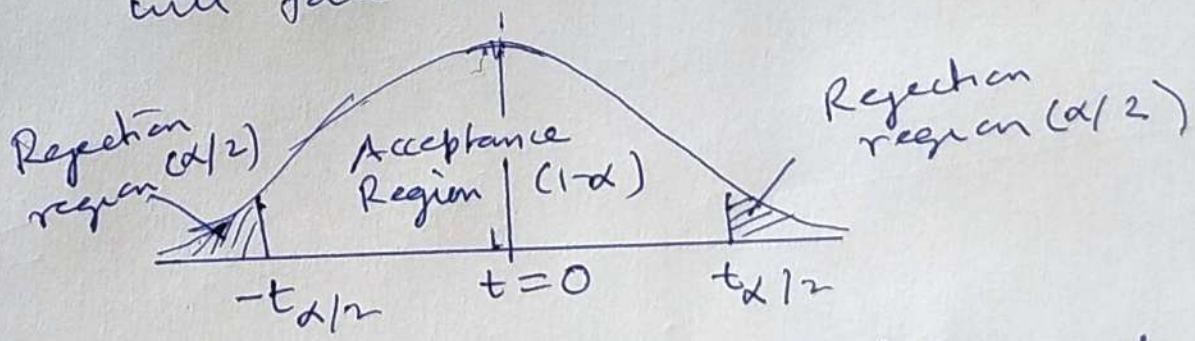
sample variance s^2

Step 3 Choose level of significance

e.g 5% (0.05) or 1% (0.01)

Step 4 Choose rejection (or critical) region for the test.

choose a set of possible test statistic values s.t. if H_0 is true, then the probability that the value of test statistic will fall in the rejection region is α.



Critical values of t: The critical (or significant) values of t at level of α and d.f ν for two-tailed test

$P(|t| > t_\nu(\frac{\alpha}{2})) = \alpha$; $P(|t| \le t_\nu(\frac{\alpha}{2})) = 1-\alpha$

Ex A random is drawn from the population with mean $\mu = 100$ Find 95% confidence limits ~~for~~ within which the mean value of sample of 10 will lie.

x_1	x_2	x_3	x_4	x_5	x_6
70	120	110	101	88	83

x_7	x_8	x_9	x_{10}
95	98	107	100.

Soln. $\bar{x} = \frac{972}{10} = 97.2$

$s^2 = 203.73$

$$|t| = \frac{|197.2 - 100|}{\sqrt{203.7310}} = 0.62$$

Two tail test

$P\left(t > t_9\left(0.\frac{05}{2}\right)\right) + P\left(t < -t_9\left(0.\frac{05}{2}\right)\right) = 0.05$

or $P\left(t > t_9\left(\frac{.05}{2}\right)\right) = 0.025$

Therefore $t_9\left(0.\frac{05}{2}\right) = 2.262$ for two-tailed test

Confidence Interval: $\left(\bar{x} \pm t_9\left(0.\frac{05}{2}\right) s/\sqrt{n}\right)$

$= (97.2 \pm 2.262 \times 4.514)$

$$P\left(t > t_{\nu,\frac{\alpha}{2}}\right) + P\left(t < -t_{\nu,\left(\frac{\alpha}{2}\right)}\right) = \alpha$$

or $2P\left(t > t_{\nu,\left(\frac{\alpha}{2}\right)}\right) = \alpha.$

or $P\left(t > t_{\nu,\left(\frac{\alpha}{2}\right)}\right) = \frac{\alpha}{2}$

<u>Ex.</u>

Single - tail test

$t_8(0.1) = 1.397$	$t_8(0.05) = 1.860$
$t_9(0.1) = 1.383$	$t_9(0.05) = 1.833$
$t_{10}(0.1) = 1.372$	$t_{10}(0.05) = 1.812$

<u>Ex</u> A machinist is requd to make some parts with diameter 0.700 inch.

Random sample of 10 parts has $\bar{x} = 0.742$ inch

$s = 0.040$ inch

Test the statistic if the work is meeting the specifications

<u>Soln</u> $H_0 : \mu = 0.700$ inch

$H_1 : \mu \neq 0.700$ inch

$$t = \frac{\bar{x} - \mu}{\sqrt{s^2/n}} = \frac{(0.742 - 0.700)\sqrt{9}}{0.040}$$

$$= \underline{3.15 \text{ (Calculated Value)}}$$

$t_{(n-1)} = t_9 = t_0$ (say) $= 1.383$ at 5% level of significance

When <u>population variance σ_X^2 and σ_Y^2</u> are unknown.

(1) Use sample estimates in place of population variances and then we use t-distribution in place of normal distn.

(2) Further if we assume $\sigma_X^2 = \sigma_Y^2 = \sigma^2$ then the two popn. are approximately normal. The common variance σ^2 is estimated as

$$S_p^2 = \frac{(n_1-1)S_X^2 + (n_2-1)S_Y^2}{(n_1+n_2-2)}$$

If null hypothesis

$$H_0 : \mu_X - \mu_Y = d_0$$

then the statistic

$$T = \frac{\bar{X} - \bar{Y} - d_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\sim t_{n_1+n_2-2}$$